Budli POC

About

Budli is a ReCommerce platform in India focused on enabling the responsible resale, refurbishment, and redistribution of used electronic devices, including smartphones, tablets, and laptops. Through a simple and transparent process—instant price discovery, free nationwide logistics, and prompt payment—Budli allows individuals and businesses to resell used gadgets in an environmentally and socially responsible manner while extending the usable life of electronic devices.

By selling certified pre-owned refurbished gadgets at accessible price points, backed by warranty and guaranteed buyback, Budli helps bridge the digital divide in India by expanding access to advanced technology for a wider segment of society. This model reduces electronic waste, promotes circular use of devices, and enables customers to participate in social impact initiatives, including donating proceeds from device sales to charitable causes.

This Proof of Concept (POC) explores how AI-assisted pricing and sell-through velocity intelligence can enhance Budli's internal pricing and decision-making workflows. The POC will assess whether combining structured, .csv-driven processes with external market signals and explainable AI-generated reasoning can improve pricing consistency, decision speed, and confidence—while maintaining full human control over final outcomes. The results will help determine the feasibility and future value of introducing scalable, explainable pricing intelligence within Budli's broader ReCommerce platform.

Objective

The goal of this 4 week Proof of Concept (POC) is to validate the technical feasibility, business value, and explainability of an AI-assisted pricing and velocity intelligence workflow that helps pricing managers evaluate selling price recommendations and expected velocity for refurbished smartphone models using structured .csv inputs and externally sourced market data.

This POC deliberately constrains scope to:

- Text-based inputs and outputs only
- .csv as the primary interaction artifact
- Up to 10 customer-defined device models per run

- Up to two external data sources selected for market scraping

The POC evaluates whether a hybrid rule-based + AI-assisted explanation approach can improve pricing consistency, speed of analysis, transparency, and learning — without automating final decisions, which remain human-approved.

Success is defined by clear feasibility, perceived value to pricing managers, explainability, and a credible path toward revenue-generating deployment, not by production readiness.

Project Success Criteria

1. .csv-Driven Pricing & Velocity Intelligence Workflow

A structured workflow that accepts a customer-provided .csv file defining multiple device models, enriches it with market signals, and returns a completed .csv output containing pricing and velocity recommendations with explanations.

Expected Capabilities

Input Handling (.csv → Structured Understanding)

- Accept an .csv file containing up to 10 device models, each defined by:
  - Brand
  - Model
  - RAM
  - Storage
  - Network type
  - Condition tier
  - Warranty option (if applicable)
- Validate completeness and consistency of input fields.
- Normalize model identifiers to ensure consistent downstream matching. normalization will rely on customer-provided definitions,

External Data Sourcing (Controlled Scope)

- Select up to three external data sources for scraping (e.g., key competitors or marketplaces). Avahi to confirm we can scrape this source. Note, this is subject to change as the website structure changes. The intention of the scrapping is to get information related to smartphone popularity and price trends to perform the analysis, customer has already set a pool of sources, we are to define the three ones that will be used. ~Three additional datasource for velocity will be used (and three for price)

- Extract textual and numeric market signals required for:
  - Price benchmarking
  - Demand/popularity inference
  - Supply and availability context
- Scraping logic is evaluated for:
  - Feasibility: The selected sources can be accessed using standard, open-source scraping tools and common extraction techniques, without requiring paid services, proprietary tooling, or complex engineering effort.
  - Stability: Sources are sufficiently stable at the time of execution to support a one-time data extraction for this assessment, with no assumption of ongoing scraping or future maintenance.
  - Repeatability: The scraping approach is documented and can be re-executed in the future.
  - Cost awareness: The solution relies only on open-source tools and existing capabilities, introducing no additional licensing or service costs.

Pricing Recommendation Logic

- Generate a recommended selling price per model based on customer-provided pricing samples, market reference prices, and agreed pricing/positioning logic (e.g., competitive benchmarks, condition-based adjustments, and demand signals) using:
  - Market reference prices
  - Relative positioning logic (e.g., competitive discount/premium)
  - Warranty and condition adjustments
  - Age and lifecycle signals
- Output pricing as:
  - A single recommended value
  - An optional confidence band (range)

Velocity Estimation

- Estimate expected selling velocity (note, from separate data sources than pricing) based on:
  - Market demand signals
  - Model popularity indicators
  - Competitive availability
- Velocity may be expressed as:
  - Categorical (Fast / Medium / Slow, thresholds to be agreed with customer), or
  - Time-bound probability buckets (POC-level fidelity)

Explainability (Mandatory)

For each model, the system must produce a human-readable explanation describing:

- Why the recommended price was chosen
- Which market signals mattered most
- Why the expected velocity category was assigned
- Any notable risks or uncertainty flags

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2. Human-in-the-Loop Review & Feedback Capture

A core success criterion is the explicit inclusion of human judgment, ensuring trust, learning, and future adoption.

Expected Capabilities

Human Review

- The output .csv contains:
  - Recommended price
  - Velocity estimate
  - Explanation
- A pricing manager reviews and may:
  - Accept the recommendation as-is
  - Adjust price up/down
  - Override velocity classification

Feedback Re-Ingestion

- The system accepts a re-uploaded .csv containing:
  - Final human-approved price
  - Acceptance or adjustment indicators
  - Optional qualitative feedback notes
- Feedback is stored in a Knowledge Base (KB).

Knowledge Base Usage

- On subsequent uploads:
  - The system checks historical feedback
  - Flags similar models or past decisions
  - Surfaces learned patterns (e.g., consistent over-pricing for a brand/condition)

Learning is directional and relative, not dependent on sensitive margin or transaction data.

UI / Interaction Model (Lightweight Workflow)

The interaction model is intentionally lightweight and file-based. The user uploads a CSV of products. The system processes the file and returns an enriched CSV for download. The user performs review and edits directly in the same CSV and re-uploads it.

No additional UI workflows are required beyond upload/download and status indicators (e.g., “Processed”, “Feedback received”). The re-uploaded CSV serves as both the feedback mechanism and the source of truth for human-approved decisions.

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3. Explainability, Trust & Professional Usability

The POC explicitly evaluates whether outputs are usable by pricing professionals, not data scientists.

Evaluation Criteria

- A pricing manager can answer:
  - “Why is this priced this way?”
  - “Why is this expected to sell faster/slower?”
  - “What risk should I be aware of?”

Success requires that explanations support decisions, not merely justify algorithms.

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4. High-Level Workflow

Input
.csv file with up to 10 customer-defined device models

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Step 1 – Input Normalization
Validate and structure device definitions

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Step 2 – Market Signal Collection
Scrape and extract required external pricing and demand signals (≤ 2 sources)

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Step 3 – Pricing & Velocity Computation
Apply rule-based logic and heuristics

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Step 4 – Explanation Generation
Produce human-readable reasoning

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Step 5 – .csv Output Generation
Return enriched .csv with:

- Price recommendation
- Velocity estimate
- Explanation

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Step 6 – Human Feedback Loop
User uploads reviewed .csv → stored in KB

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Step 7 – Knowledge-Aware Future Runs
Historical feedback informs subsequent recommendations

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5. Deliverables

By the end of the POC, the following artifacts will be delivered:

- Working .csv-based pricing & velocity POC
- Defined scraping scope and data source feasibility assessment
- Sample input and output .csv templates
- Explanation samples across multiple models
- Knowledge Base schema for feedback storage
- Summary report

Final Outcome & Success Definition

By the end of this POC, the customer will have:

- Validated whether AI-assisted pricing and velocity recommendations are feasible using:
  - Limited external data sources
  - .csv-first workflows

    - Human-approved outputs
- Evidence that:
    - Pricing decisions can be made faster
    - Market reasoning is clearer and more consistent
    - Learning from feedback is technically viable

Success is not defined by automation or production deployment, but by:

- Technical feasibility
- Professional trust
- Explainability
- And a credible path toward business value and monetization

Scope of Work

This engagement is a 4-week Proof of Concept (POC) designed to evaluate the technical feasibility, usability, and business value of an AI-assisted workflow for pricing recommendation and sell-through velocity estimation for refurbished smartphone models.

The POC focuses on validating whether a structured, .csv-driven, human-in-the-loop workflow—augmented with external market signals and AI-generated explanations—can improve pricing decision quality, speed, and consistency, while maintaining full human control over final outcomes.

The scope is intentionally constrained to ensure clarity, cost control, and decision-oriented outcomes.

Scope Constraints

The POC is explicitly limited to:

- Text-based inputs and outputs only (no UI, no dashboards)
- .csv as the primary interaction artifact
- Up to 10 customer-defined device models per run
- Selection of up to two external data sources for market signal extraction
- One-time data scraping only (no live or recurring scraping)
- Human-approved pricing and velocity outputs (no automation)

This engagement emphasizes experimentation, explainability, and learning validation, not production deployment.

Pre-Kickoff

Activities

- Alignment session with customer stakeholders to confirm:
  - Business objectives and POC intent
  - .csv-based workflow expectations
- Confirmation of AWS environment access Avahi Onboarding.
- Identification and approval of:
  - Input CSV format and required fields will be standardized, and the customer will provide a canonical device naming reference along with representative samples of all naming variants used in their systems, to reduce normalization ambiguity.Up to two external market data sources to be scraped
- Establish communication channels (Slack / Teams) and shared documentation repository.

Deliverables

- AWS access confirmed.
- Approved .csv input template.
- Approved list of external data sources.
- Communication channels and shared documentation space established.

Phase 1: Discovery and Planning

Duration: Week 1

Activities

- Formal project kickoff with customer stakeholders.
- Review and confirmation of Project Success Criteria, including:
  - Pricing recommendation quality
  - Velocity estimation usefulness
  - Explanation clarity
  - Human review and feedback loop viability
- Validate scope boundaries and exclusions (no live scraping, no UI, no automation).
- Review structure and variability of .csv inputs.
- Define evaluation framework for:
  - Pricing logic outputs

    - Velocity classification
    - Explanation quality
- Align on acceptance checkpoints and final readout format.

Deliverables

- Finalized project plan and 4-week timeline.
- Confirmed success criteria and evaluation metrics.
- Approved .csv input samples and model definitions.

Phase 2: Workflow Design and Development

Duration: Weeks 2–3

Activities

.csv Input Processing

- Implement structured ingestion and validation of .csv files containing:
    - Device specifications
    - Condition tiers
    - Warranty attributes (if applicable)
- Normalize device identifiers for consistent downstream processing.

One-Time Market Data Extraction

- Execute one-time scraping of approved external data sources.
- Extract required pricing and demand-related signals.
- Store extracted data for repeatable POC execution (no re-scraping).

Pricing Recommendation Logic

- Implement rule-based pricing logic using:
    - Market reference prices
    - Relative positioning heuristics
    - Condition and warranty adjustments
    - Product age and lifecycle indicators
- Produce a recommended selling price per model.

Velocity Estimation

- Derive expected sell-through velocity using available demand and market signals.
- Express velocity as:

  - Categorical (Fast / Medium / Slow), or
  - Time-bucketed estimates (POC-level fidelity).

Explainability Generation

- Generate human-readable explanations for each model describing:
  - Key drivers behind pricing
  - Drivers behind velocity classification
  - Risk or uncertainty indicators
- Ensure explanations are suitable for pricing professionals, not technical users.

Feedback Loop Design

- Design .csv-based feedback capture, allowing users to:
  - Accept or adjust recommendations
  - Upload reviewed .csv outputs
- Persist feedback into a lightweight Knowledge Base for future reference.

Deliverables

- Functional POC workflow covering:
  - .csv input ingestion
  - One-time market data usage
  - Pricing and velocity outputs
  - Explanation generation
  - Feedback re-ingestion
- Sample enriched .csv output files.
- Initial documentation of pricing and velocity logic.

Phase 3: Testing, Evaluation, and Handover

Duration: Week 4

Activities

- Execute end-to-end POC runs using agreed sample .csv inputs.
- Validate outputs against agreed success criteria:
  - Pricing relevance: Alignment of recommended prices with market reference signals and business expectations, qualitatively assessed by pricing managers.
  - Velocity usefulness: Practical value of velocity estimates for inventory, reselling, or prioritization decisions, based on stakeholder judgment and operational fit.

    - Explanation clarity: Interpretability and actionability of explanations for business users, as reviewed by pricing managers.
- Conduct qualitative review sessions with stakeholders.
- Evaluate feasibility of learning from feedback across runs.
- Prepare final POC demo and findings readout.
- Develop clear recommendations for:
    - Next-phase experimentation
    - Potential productization
    - Revenue-linked deployment paths

Deliverables

- Final end-to-end POC demonstration.
- Lightweight documentation package including:
    - Workflow and architecture overview
    - Key findings and insights
    - Clear next steps and go-forward recommendations

Service Delivery Assumptions and Exclusions

To ensure focused, timely, and controlled delivery of this 4-week Proof of Concept (POC), the following service delivery assumptions and exclusions apply.

This engagement is limited to early-stage pricing recommendation and sell-through velocity estimation for refurbished smartphone models based on structured .csv inputs and externally sourced market signals. Activities related to downstream commercial operations—such as automated price publishing, inventory optimization, procurement execution, ERP integration, or revenue forecasting—are explicitly excluded.

The POC will operate on .csv files containing up to 10 customer-defined device models per run. .csv is the sole interaction artifact for inputs, outputs, and feedback. No dashboards, standalone applications, or user interfaces are included.

The engagement includes selection of up to two external data sources for market signal extraction. All data scraping will be performed once only for POC evaluation purposes. No live, scheduled, real-time, or recurring scraping is included. Ongoing data refresh, monitoring of source availability, or long-term data maintenance is out of scope.

External data availability, structure, and accessibility are assumed to be reasonably stable at the time of scraping. Changes in source layouts, access restrictions, blocking mechanisms, or data completeness after the one-time scrape are not covered.

Pricing and velocity recommendations are advisory only. All outputs require explicit human review and approval. The POC does not include automated pricing decisions, automated purchasing decisions, or direct system actions of any kind.

Velocity estimation is intended for directional decision support (e.g., Fast / Medium / Slow or indicative time buckets). It is not a guarantee of sales performance, sell-through timing, or financial outcomes.

All development and testing will be conducted in a non-production AWS environment. Outputs are intended strictly for feasibility, usability, and value evaluation and are not suitable for live commercial use.

Evaluation will rely on pre-defined rules, heuristics, and AI-assisted explanation generation. No custom model training, fine-tuning, supervised learning, reinforcement learning, or statistically validated forecasting models are included.

The feedback and learning loop is limited to capturing and referencing historical human adjustments. It is intended to demonstrate learning feasibility, not to deliver a production-grade adaptive pricing system.

No integrations with customer production systems are included, including but not limited to:

- ERP systems
- Inventory or warehouse management systems
- E-commerce platforms
- Pricing engines
- Procurement or vendor management tools

No advanced analytics dashboards, long-term reporting systems, business intelligence tooling, or visualization layers are included.

No performance guarantees, scalability commitments, load testing, latency SLAs, or cost optimization guarantees are provided as part of this POC.

No compliance, legal, regulatory, or audit activities are included. This includes (but is not limited to) financial compliance, consumer protection compliance, pricing regulation audits, security certifications, penetration testing, or formal risk assessments.

No UI/UX design work, frontend development, or mobile application development is included beyond generating structured .csv outputs.

Ongoing support, maintenance, enhancements, or data refresh beyond the 4-week POC duration are not included unless defined in a separate, mutually agreed follow-on engagement.

Any feature, capability, activity, or deliverable not explicitly stated in the Scope of Work is considered excluded unless formally added through a written change request approved by both parties.